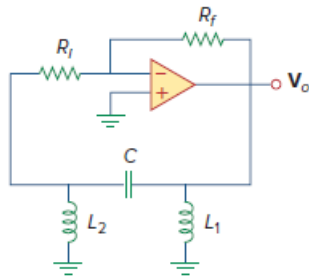


Problem

Figure 10.136 shows a *Hartley oscillator*. Show that the frequency of oscillation is

$$f_o = \frac{1}{2\pi \sqrt{C(L_1 + L_2)}}$$

Figure 10.136: A Hartley oscillator



Step-by-step solution

Step 1 of 4

Draw the following circuit diagram:

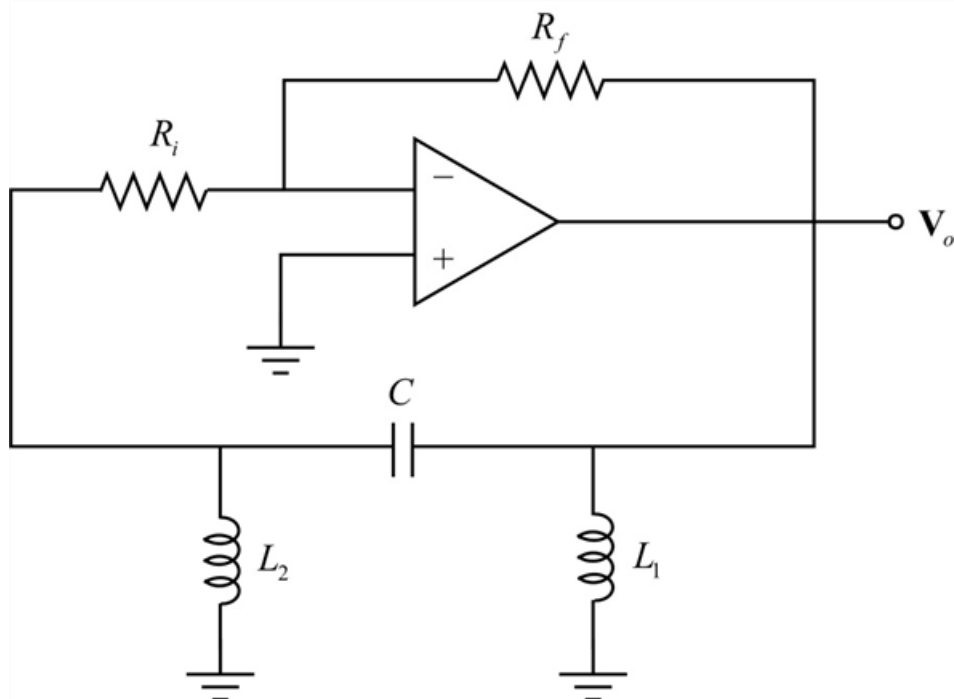


Figure 1

Step 2 of 4

Draw the feedback part of the circuit.

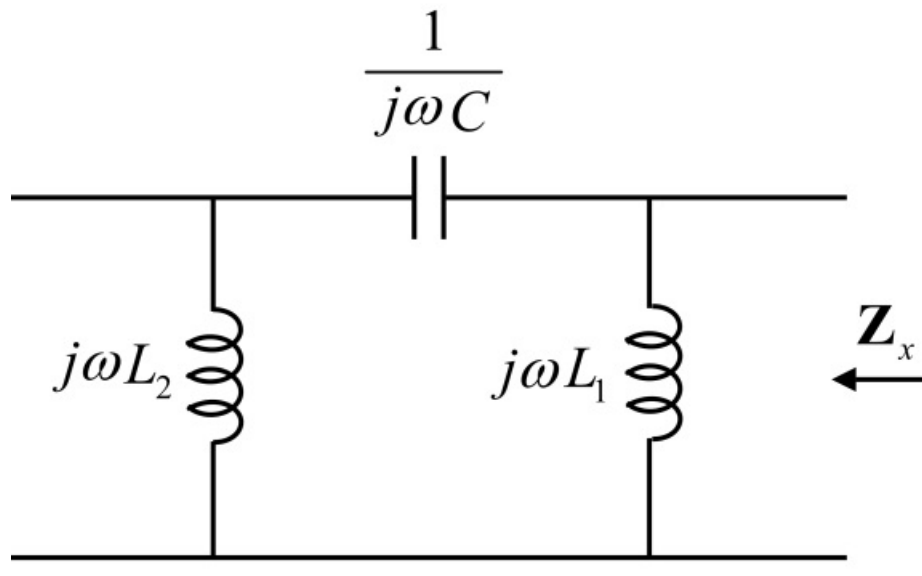


Figure 2

Step 3 of 4

Calculate the impedance $\mathbf{Z}_x$ for the circuit in Figure 2.

$$\begin{aligned}
 \mathbf{Z}_x &= \left(\frac{1}{j\omega C} + j\omega L_2 \right) \parallel (j\omega L_1) \\
 &= \left(\frac{-j}{\omega C} + j\omega L_2 \right) \parallel (j\omega L_1) \\
 &= \frac{\left(\frac{-j}{\omega C} + j\omega L_2 \right) (j\omega L_1)}{\left(\frac{-j}{\omega C} + j\omega L_2 \right) + (j\omega L_1)} \\
 &= \frac{\left[\frac{L_1}{C} - \omega^2 L_1 L_2 \right]}{j \left[\frac{-1}{\omega C} + \omega L_1 + \omega L_2 \right]} \\
 &= \frac{\left[\frac{L_1 - \omega^2 L_1 L_2 C}{C} \right]}{j \left[\frac{-1 + \omega^2 C L_1 + \omega^2 C L_2}{\omega C} \right]} \\
 &= \frac{\omega L_1 (1 - \omega^2 L_2 C)}{j [\omega^2 C (L_1 + L_2) - 1]}
 \end{aligned}$$

Step 4 of 4

The impedance $\mathbf{Z}_x$ is purely real, when the imaginary part equal to zero.

$$\omega_o^2 C(L_1 + L_2) - 1 = 0$$

$$\omega_o^2 C(L_1 + L_2) = 1$$

$$\omega_o^2 = \frac{1}{C(L_1 + L_2)}$$

$$(2\pi f_o)^2 = \frac{1}{C(L_1 + L_2)}$$

$$2\pi f_o = \frac{1}{\sqrt{C(L_1 + L_2)}}$$

$$f_o = \frac{1}{2\pi\sqrt{C(L_1 + L_2)}}$$

Therefore, the frequency $f_o = \frac{1}{2\pi\sqrt{C(L_1 + L_2)}}$ is proved.